DEVANSHI VASHISTHA

Data Scientist ◇ Researcher ◇ AI Reviewer
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PROFILE

I build machine learning that holds up after the demo, then go looking for where it breaks. **2+ years of industry experience**: shipped a real-time computer vision system at **99% accuracy on AWS** and was promoted to team lead over 4 interns. As a researcher, I published peer-reviewed CNN work at IEEE and trained a calibrated classifier on **45,152 ECGs** that catches **88.8% of serious cardiac cases**, then audited who it fails. As a reviewer, I grade frontier LLM outputs for model training at Mercor. Bring me a messy data problem and I am happy to talk it through.

TECHNICAL SKILLS

Languages	Python, SQL, R
Machine Learning	scikit-learn, XGBoost, LightGBM, TensorFlow, Keras, CNNs, NLP, feature engineering
Research & Evaluation	SHAP, hypothesis testing, effect sizes, calibration, fairness audits, cross-validation, LLM output evaluation, rubric-based grading
Vision & Signals	OpenCV, NeuroKit2, image classification, object detection, ECG signal processing
Deployment & Tools	AWS, Docker, Flask REST APIs, Git, GitHub, Postman, Jupyter
Data & BI	pandas, NumPy, Matplotlib, Tableau, MS Excel
Currently learning	PyTorch, retrieval-augmented generation (RAG) and LLM evaluation pipelines

PROFESSIONAL EXPERIENCE

AI / ML Software Developer, promoted to Team Lead Jun 2024 – Jul 2025
Agsure Innovations Private Limited *Delhi, India*

- **Situation:** grain quality was graded by eye. **Action:** built and tuned computer vision models for automated grading. **Result:** a real-time production system at **99% accuracy**.
- Deployed the service on **AWS** and moved TensorFlow / Keras models into production pipelines behind **Flask REST APIs**, tested with Postman.
- **Promoted to team lead, guiding 4 interns** through experiments, code reviews and deployment, and explained model decisions to non-technical colleagues.

Freelance AI Evaluation Specialist (AI Training) Jul 2026 – Present
Mercor *Remote, part time*

- Grade LLM outputs against a fixed **eight-category rubric**, scoring each criterion separately so every judgement can be used directly as a training signal.
- Review content across ESG, cybersecurity, healthcare and investment topics, checking labels against reference answers to stay consistent batch to batch.
- Tag "AI-isms" (padded phrasing, empty hedging, template structure) and write prompt guidance that pushes models toward specific, natural answers.

Sales Development Representative Intern Dec 2023 – Jun 2024
Recruit CRM *Remote*

- Generated leads and ran cold outreach to recruitment agencies across North America for a B2B SaaS product.
- Set up and coordinated live product demos with regional sales leaders.
- First point of contact for prospects across North America, Europe and Asia; qualified buying interest and kept records in HubSpot CRM.

AI / ML Trainee (Industrial Training) Jun 2023 – Jul 2023
Jaypee University of Information Technology *Solan, India*

- Completed hands-on training in Python, classical machine learning and neural networks.
- Implemented data preprocessing pipelines: cleaning, encoding, scaling and train / test splits.
- Built and compared baseline ML models as part of the training coursework.

Data Scientist (Apprenticeship) Jul 2022 – Oct 2022
IBM *Remote*

- Applied machine learning techniques to several datasets to solve predictive problems.
- Contributed to an applied research project focused on predictive modelling.
- Explored, cleaned and visualised data in Python before modelling, and reported results to project mentors.

CASE STUDIES

01 Arrhythmia Triage from a 10-second ECG (MSc Dissertation) *Jun 2026 – Sep 2026 · with Uniworx*

- **Problem:** flag serious cardiac abnormalities from a single Lead II window across 45,152 PhysioNet ECGs with severe class imbalance.
- **Approach:** NeuroKit2 pipeline extracting 21 HRV, rhythm and morphology features; removed SMOTE after finding 41% of training rows were synthetic; cut collinearity (max VIF 236.9 to 12.6) before SHAP; compared RF, XGBoost and LightGBM over 16 configurations and 30 paired runs with Holm-corrected tests.
- **Outcome:** calibrated LightGBM catching **88.8% of serious cases** (95% CI 87.6–90.1%) at **86.9% accuracy**, macro AUC 0.968; XGBoost reached 0.883 recall. The fairness audit exposed a recall gap across age bands (0.29 to 0.98), reported as a key limitation.

02 Making NDA Risk Visible: Project:Hack27, Rolls-Royce L&D Centre, Derby *Apr 2026*

- Two-day hackathon with team Collective Voice: analysed 1,656 anonymised survey responses; built a **Sector Risk Lens** scoring 20+ sectors and a logistic regression model of career harm after reporting, explained in plain English for policymakers.

03 Does a Deeper CNN Find Tumours Better? (IEEE ICIIP 2023) *Published research*

- Compared CNNs of increasing hidden-layer depth on brain MRI across four tumour classes; the best model reached **95% accuracy**, published at IEEE ICIIP 2023.

04 Applicant Tracking System *NLP*

- Placement staff matched resumes to roles by hand; built an NLP system that parses resumes and ranks candidates, improving applicant quality by **85%**.

05 Driver Drowsiness Detection *Real-time computer vision*

- Tracked eye state from live video with OpenCV and a Keras classifier to trigger fatigue alerts, reaching **91% accuracy** in real time.

EDUCATION

MSc Data Science 2025 – 2026
University of Leicester, United Kingdom *Expected: Distinction*

BTech Computer Science and Engineering 2020 – 2024
Jaypee University of Information Technology, India *CGPA: 8.9 / 10*

RESEARCH & PUBLICATIONS

D. Vashistha, S. Goel, S. Jain, "Impact of Convolutional Neural Network Hidden Layers on Brain Tumor Detection," *Seventh International Conference on Image Information Processing (ICIIP)*, IEEE, 2023. DOI: 10.1109/ICIIP61524.2023.10537735

CERTIFICATIONS

IBM: Deep Learning using TensorFlow • Google: Introduction to Generative AI • Infosys: Data Visualization using Python • Languify: Artificial Intelligence with Python • JP Morgan: Excel Skills Job Simulation